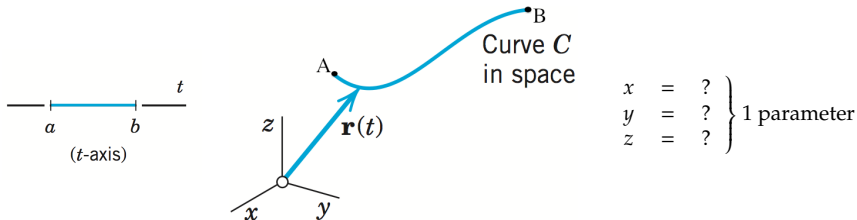


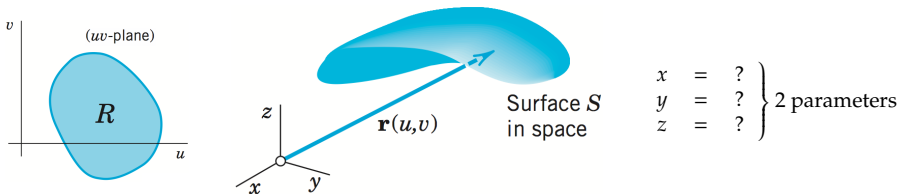
Topic: Surfaces & Surface Integrals

I. Surface Parameterization

Line Parameterization:



Surface Parameterization:



Ex: Find a parameterization of the following surfaces:

(a) $15x + 2z = 12$ (plane extending in the y direction)

Note: y is independent of x and z , so it should be one of the parameters.

We are free to choose x or z as the other parameter.

$$\begin{array}{llll} x = x & & x = v & \\ y = y & \xRightarrow[u=v]{u=y} & y = u & \implies \mathbf{r}(u, v) = \left[v, u, 6 - \frac{15}{2}v \right] \\ z = \frac{1}{2}(12 - 15x) & & z = 6 - \frac{15}{2}v & -\infty \leq u \leq \infty, -\infty \leq v \leq \infty \end{array}$$

(b) $(x + 2)^2 + z^2 = 16$ (cylinder extending in the y direction)

Note: Again, since y is independent of x and z , so it should be one of the parameters.

Since x and z have a circular relationship (radius 4), so let's consider polar coordinates:

$$\frac{(x+2)^2}{16} + \frac{z^2}{16} = 1 \implies \left(\frac{x+2}{4} \right)^2 + \left(\frac{z}{4} \right)^2 = 1$$
$$\begin{array}{ccc} \uparrow & & \uparrow \\ \cos \theta & & \sin \theta \end{array}$$

$$\begin{array}{llll} x = 4 \cos \theta - 2 & & x = 4 \cos v - 2 & \\ y = y & \xRightarrow[u=v]{u=y} & y = u & \implies \mathbf{r}(u, v) = [4 \cos v - 2, u, 4 \sin v] \\ z = 4 \sin \theta & & z = 4 \sin v & -\infty \leq u \leq \infty, 0 \leq v \leq 2\pi \end{array}$$

(c) $x^2 + y^2 - z^2 = 0$ in the first octant and $0 \leq z \leq 6$ (cone extending in the z direction)

Note: Writing the above we have: $x^2 + y^2 = z^2$. This appears as a circle of radius z .

Writing this circle in polar coordinates would be:

$$x^2 + y^2 = z^2 \implies \left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 = 1$$

$\uparrow \qquad \uparrow$
 $\cos \theta \quad \sin \theta$

$$\begin{array}{llll} x = z \cos \theta & & x = u \cos v & \\ y = z \sin \theta & \xRightarrow[u=v]{u=z} & y = u \sin v & \implies \mathbf{r}(u, v) = [u \cos v, u \sin v, u] \\ z = z & & z = u & 0 \leq u \leq 6, 0 \leq v \leq \frac{\pi}{2} \end{array}$$

(d) $4x + 10y + 2z = 1$ where $-1 \leq y \leq 2$ and $1 \leq z \leq 5$ (plane)

Note: Though we are free to set u and v as any of the variables x, y, z , it is most convenient to use y and z since we already know their bounds.

$$\begin{array}{llll} x = \frac{1}{4}(1 - 10y - 2z) & & x = \frac{1}{4} - \frac{5}{2}u - \frac{1}{2}v & \\ y = y & \xRightarrow[v=z]{u=y} & y = u & \implies \mathbf{r}(u, v) = \left[\frac{1}{4} - \frac{5}{2}u - \frac{1}{2}v, u, v\right] \\ z = z & & z = v & -1 \leq u \leq 2, 1 \leq v \leq 5 \end{array}$$

(e) $-4x + 5y^2 + 7z^2 = 0$ in the first octant and $0 \leq z \leq 6, 1 \leq y \leq 2$ (paraboloid along x -axis)

Note: Although polar could be used in this case, the better option is usually rectangular

Solve for x and set y and z as parameters

$$\begin{array}{lll} x = \frac{5}{4}y^2 + \frac{7}{4}z^2 & & x = \frac{5}{4}u^2 + \frac{7}{4}v^2 \\ y = y & \xRightarrow[u=v]{u=y} & y = u \\ z = z & & z = v \end{array} \implies \mathbf{r}(u, v) = \left[\frac{5}{4}u^2 + \frac{7}{4}v^2, u, v \right]$$
$$1 \leq u \leq 2, 0 \leq v \leq 6$$

(f) $4x^2 + y^2 + 9z^2 = 4$ (ellipsoid) Note: Need spherical coordinates

$$\frac{4x^2}{4} + \frac{y^2}{4} + \frac{9z^2}{4} = 1 \implies \underset{\substack{\uparrow \\ \sin \phi \cos \theta}}{(x)^2} + \underset{\substack{\uparrow \\ \sin \phi \sin \theta}}{\left(\frac{y}{2}\right)^2} + \underset{\substack{\uparrow \\ \cos \phi}}{\left(\frac{3z}{2}\right)^2} = 1$$

$$\begin{array}{lll} x = \sin \phi \cos \theta & & x = \sin u \cos v \\ y = 2 \sin \phi \sin \theta & \xRightarrow[u=v]{u=\phi} & y = 2 \sin u \sin v \\ z = \frac{2}{3} \cos \phi & & z = \frac{2}{3} \cos u \end{array} \implies \mathbf{r}(u, v) = \left[\sin u \cos v, 2 \sin u \sin v, \frac{2}{3} \cos u \right]$$
$$0 \leq u \leq \pi, 0 \leq v \leq 2\pi$$

II. Surface Tangents and Normal Vectors

Let $\mathbf{r}(u, v) = [x(u, v), y(u, v), z(u, v)]$ be a parameterization of a surface, S .

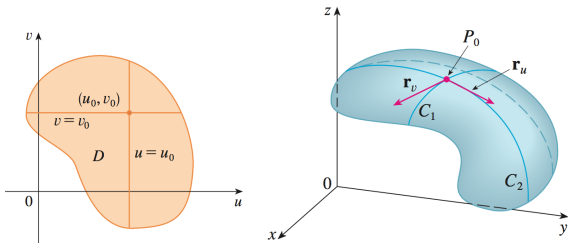
Surface Tangents:

Tangent along C_2 :

$$\mathbf{r}_u = \left[\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right]$$

Tangent along C_1 :

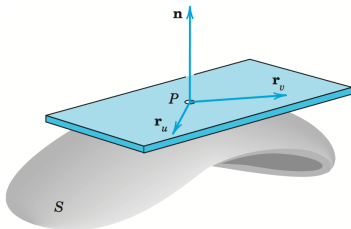
$$\mathbf{r}_v = \left[\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right]$$



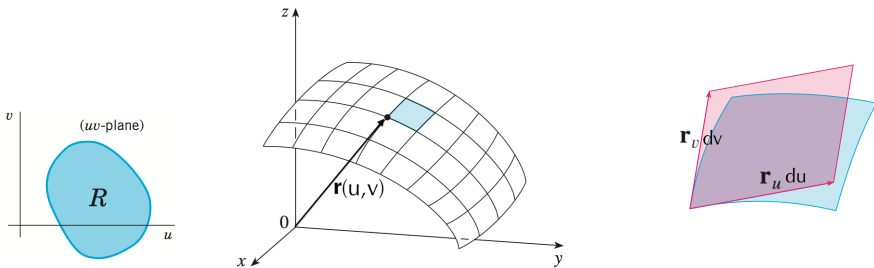
Normal:

$$\text{Normal: } \mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v$$

$$\text{Unit Normal: } \mathbf{n} = \frac{\mathbf{N}}{|\mathbf{N}|} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|}$$



Derive the integral expression for surface area of a surface S with parameterization $\mathbf{r}(u, v)$:



Area of S = Sum of many “small” areas (Scalar Quantity)

$$\Rightarrow \text{Area}(S) = \iint_S dS \quad \text{What is } dS?$$

dS represents the area of the small part of the plane approximating the patch of S (see figure)

Note that the small part is a parallelogram spanned by the vectors: $\mathbf{r}_u du$ and $\mathbf{r}_v dv$

Therefore, $dS = |\mathbf{r}_u du \times \mathbf{r}_v dv| = |\mathbf{r}_u du \times \mathbf{r}_v| dv = |\mathbf{r}_u \times \mathbf{r}_v| du dv$ and so

$$\text{Area}(S) = \iint_S dS = \iint_R |\mathbf{r}_u \times \mathbf{r}_v| du dv$$

III. Surface Integrals

Let $\mathbf{F}$ be a vector field.

Line Integral - C is a curve

$$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Surface Integral - S is a surface

$$\int_S \mathbf{F}(\mathbf{r}) \cdot d\mathbf{A} = \iint_S \mathbf{F} \cdot \mathbf{n} dA = \iint_S \mathbf{F} \cdot \frac{\mathbf{N}}{|\mathbf{N}|} |\mathbf{r}_u \times \mathbf{r}_v| du dv = \iint_R \mathbf{F}(\mathbf{r}(u,v)) \cdot \mathbf{N}(u,v) du dv$$

where $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v$ is the normal vector to the surface and $\mathbf{n} = \frac{\mathbf{N}}{|\mathbf{N}|}$ is the unit normal. R is the region in the u,v -plane (parameter plane).

IV. Flux

Flux is a very important concept when modeling physical phenomena.

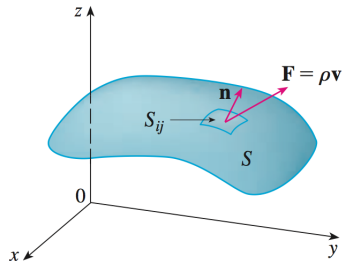
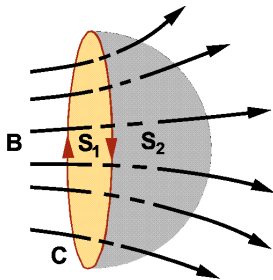
Mathematically, it is given by the surface integral

$$\text{Flux through a surface } S = \iint_S \mathbf{F} \cdot d\mathbf{S}$$

Where $\mathbf{F}$ is a vector field and $\mathbf{n}$ is the unit normal to the surface.

When dealing with flow through a surface, the vector field is typically written as: $\mathbf{F} = \rho \mathbf{v}$

where ρ is the density of the flowing substance and $\mathbf{v}$ is the velocity of the flow.



Ex: Compute the flux of water through the parabolic cylinder $S : y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 3$ if the velocity flow of water is given by $\mathbf{v} = [3z^2, 6, 6xz]$ (units: m/s). Note, the density of water is $\rho = 1 \text{ kg}/m^3$.

Set-up: we need $\mathbf{F}$, $\mathbf{r}(u, v)$, $\mathbf{r}_u$, $\mathbf{r}_v$ and $\mathbf{N}$

Since we are dealing with fluid flow, $\mathbf{F} = \rho \mathbf{v} \implies \mathbf{F} = [3z^2, 6, 6xz]$

For $\mathbf{r}(u, v)$, we need

$$\begin{aligned} x &= u \\ y &= u^2 \\ z &= v \end{aligned} \implies \mathbf{r}(u, v) = [u, u^2, v] \quad \text{where } 0 \leq u \leq 2 \text{ and } 0 \leq v \leq 3$$

$$\implies \mathbf{r}_u = [1, 2u, 0] \quad \text{and} \quad \mathbf{r}_v = [0, 0, 1]$$

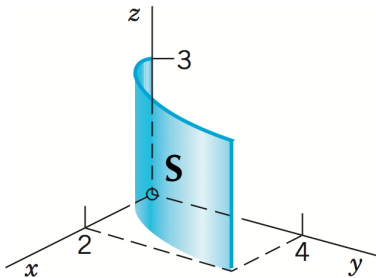
$$\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2u & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 2u & 0 \\ 0 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2u \\ 0 & 0 \end{vmatrix} \mathbf{k} = 2u \mathbf{i} - \mathbf{j} + 0 \mathbf{k} = [2u, -1, 0]$$

$$\mathbf{F}(\mathbf{r}(u, v)) = [3v^2, 6, 6uv]$$

$$\mathbf{F}(\mathbf{r}(u, v)) \cdot \mathbf{N} = [3v^2, 6, 6uv] \cdot [2u, -1, 0] = 6uv^2 - 6$$

$$\begin{aligned}
 \text{Flux} &= \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^3 \int_0^2 6uv^2 - 6 \, du \, dv \\
 &= \int_0^3 \left[6 \left(\frac{u^2}{2} \right) v^2 - 6u \right]_{u=0}^{u=2} dv \\
 &= \int_0^3 [6(2)v^2 - 6(2)] dv \\
 &= 4v^3 - 12v \Big|_0^3 = 4(27) - 12(3) = 72 \text{ kg/s}
 \end{aligned}$$

Translation: 72 kg of water pass through the surface S every second



Ex: Compute the surface integral of $\mathbf{F} = [e^y, -e^z, e^x]$ over the surface $S : x^2 + y^2 = 25$ for $x \geq 0$, $y \geq 0$, and $0 \leq z \leq 2$

$$x = 5 \cos u$$

$$y = 5 \sin u \quad \Rightarrow \quad \mathbf{r}(u, v) = [5 \cos u, 5 \sin u, v] \quad \text{where} \quad 0 \leq u \leq \frac{\pi}{2} \text{ and } 0 \leq v \leq 2$$

$$z = v$$

$$\Rightarrow \quad \mathbf{r}_u = [-5 \sin u, 5 \cos u, 0] \quad \text{and} \quad \mathbf{r}_v = [0, 0, 1]$$

$$\begin{aligned} \mathbf{N} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -5 \sin u & 5 \cos u & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 5 \cos u & 0 \\ 0 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -5 \sin u & 0 \\ 0 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -5 \sin u & 5 \cos u \\ 0 & 0 \end{vmatrix} \mathbf{k} \\ &= 5 \cos u \mathbf{i} + 5 \sin u \mathbf{j} + 0 \mathbf{k} = [5 \cos u, 5 \sin u, 0] \end{aligned}$$

$$\mathbf{F}(\mathbf{r}(u, v)) = [e^{5 \sin u}, -e^v, e^{5 \cos u}]$$

$$\mathbf{F}(\mathbf{r}(u, v)) \cdot \mathbf{N} = [e^{5 \sin u}, -e^v, e^{5 \cos u}] \cdot [5 \cos u, 5 \sin u, 0] = 5e^{5 \sin u} \cos u - 5e^v \sin u$$

$$\begin{aligned}
\iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \int_0^2 \int_0^{\pi/2} 5e^{5\sin u} \cos u - 5e^v \sin u \, du \, dv & w &= 5 \sin u \\
&= \int_0^2 \int_{u=0}^{u=\pi/2} e^w - 5e^v \sin u \, du \, dv & dw &= 5 \cos u \, du \\
&= \int_0^2 [e^w + 5e^v \cos u]_{u=0}^{u=\pi/2} \, dv \\
&= \int_0^2 [e^{5\sin u} + 5e^v \cos u]_{u=0}^{u=\pi/2} \, dv \\
&= \int_0^2 [e^{5\sin(\pi/2)} + 5e^v \cos(\pi/2)] - [e^{5\sin(0)} + 5e^v \cos(0)] \, dv \\
&= \int_0^2 e^5 - 1 - 5e^v \, dv \\
&= v(e^5 - 1) - 5e^v \Big|_0^2 \\
&= [2(e^5 - 1) - 5e^2] - [-5] = 2(e^5 - 1) - 5e^2 + 5
\end{aligned}$$

V. Other Surface Integrals

Let g be a scalar-valued function.

Line Integral - C is a curve

$$\int_C g \, ds = \int_a^b g(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt$$

Surface Integral - S is a surface

$$\int_S g \, dA = \iint_R g(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, du \, dv = \iint_R g(\mathbf{r}(u, v)) |\mathbf{N}(u, v)| \, du \, dv$$

R is the region in the u, v -plane (parameter plane).

Ex: Compute the surface integral $\iint_S (1 + 9xz)^{3/2} dA$

where S is given by the parameterization: $\mathbf{r}(u, v) = [u, v, u^3]$, $0 \leq u \leq 1$, $-2 \leq v \leq 2$

$$\Rightarrow \quad \mathbf{r}_u = [1, 0, 3u^2] \quad \text{and} \quad \mathbf{r}_v = [0, 1, 0]$$

$$\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 3u^2 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 3u^2 \\ 1 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 3u^2 \\ 0 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \mathbf{k} = -3u^2 \mathbf{i} - 0 \mathbf{j} + \mathbf{k} = [-3u^2, 0, 1]$$

$$g(\mathbf{r}(u, v)) = (1 + 9u^4)^{3/2} \quad \text{and} \quad |\mathbf{N}| = (9u^4 + 1)^{1/2}$$

$$g(\mathbf{r}(u, v)) |\mathbf{N}| = (1 + 9u^4)^2 = 1 + 18u^4 + 81u^8$$

$$\begin{aligned} \int_S (1 + 9xz)^{3/2} dA &= \int_{-2}^2 \int_0^1 1 + 18u^4 + 81u^8 \, du dv \\ &= \int_{-2}^2 \left[u + \frac{18}{5} u^5 + \frac{81}{9} u^9 \right]_{u=0}^{u=1} dv \\ &= \int_{-2}^2 1 + \frac{18}{5} + 9 \, dv = \int_{-2}^2 \frac{68}{5} \, dv = (4) \frac{68}{5} = 272 \end{aligned}$$